N-Way vs. Recursive Bisection: A General Architectural Comparison

B.5 — Algorithm Track

Gio Della-Libera
Apportionment Research Group

May 2026

Abstract

Congressional redistricting algorithms divide into two broad families: those that partition a state graph directly into k parts (*n-way*), and those that recursively bisect into two halves until k leaves are reached (*recursive bisection*). Prior work (B.3, B.4) established equivalence under edge weighting for the specific case of congressional maps. This paper generalises to arbitrary chambers, district counts, and graph resolutions. We find that recursive bisection consistently matches or outperforms n-way partitioning on compactness across all tested configurations, while n-way is faster by $O(\log k)$ on large instances. For redistricting purposes the compactness advantage of recursive bisection is decisive: the runtime difference is below measurement noise on modern hardware.

1 Introduction

Graph partitioning for redistricting admits two natural strategies. *N-way partitioning* (direct k-way) calls a partitioner once with target k, allowing the algorithm global visibility of all k parts simultaneously. *Recursive bisection* (RB) splits the graph into two halves, then recursively applies the same procedure to each half until exactly k leaves exist.

The tradeoff is well-studied in the scientific-computing literature [Karypis and Kumar, 1998] but has not been evaluated in the redistricting context across the full range of k values, geographic resolutions, and balance constraints encountered in practice. B.3 showed single-objective equivalence for congressional maps; B.4 extended this to edge-weighted formulations. Here we generalise across chambers (congressional, state house, state senate) and resolutions (tract, block group, block).

2 Methodology

2.1 Test configurations

We evaluate 435 congressional districts (50 states), 99 state senate chambers, and 193 state house chambers from the 2020 Census. Resolution varies from tract (∼74,000 units nationally) to block group (∼239,000).

2.2 Metrics

Compactness. Mean Polsby–Popper score $\overline{PP}$ per chamber run; lower is less compact.

Runtime. Wall-clock time on a 16-core workstation, averaged over 10 seeds.

Balance deviation. Maximum absolute deviation from the ideal population per district, as a fraction of the ideal.

2.3 Implementation

Both strategies use the same METIS partitioner (C FFI, `c-ffi` engine) with identical hyperparameters: `ufactor=5`, `niter=100`, `seed=42`.

3 Results

Table 1: Mean Polsby–Popper by strategy and chamber type (2020 Census)

Strategy	Congressional	State Senate	State House
Recursive bisection	0.361	0.344	0.329
N-way (direct k)	0.357	0.341	0.325
Difference	+0.004	+0.003	+0.004

Recursive bisection outperforms n-way by approximately +0.004 mean PP across all chamber types. The advantage narrows for prime k values (PA k=17, NV k=42) where recursive bisection must use asymmetric binary fallbacks.

Runtime favours n-way by 18–34% on large state house chambers ($k > 80$) but the absolute difference is under 200 ms per state.

4 Discussion

The compactness advantage of recursive bisection is attributable to the hierarchical constraint structure: at each level, the bisector optimises balance within a contiguous subgraph rather than globally across all k parts simultaneously. This locality produces more geographically coherent districts.

For redistricting applications we recommend recursive bisection as the default. The runtime penalty is negligible; the compactness benefit is consistent and statistically significant ($p < 0.001$ by paired t-test across 50 states).

5 Conclusion

Across all tested configurations — congressional, state senate, and state house chambers at tract and block-group resolution — recursive bisection matches or outperforms direct n-way partitioning on compactness at negligible runtime cost. These results generalise the equivalence result of B.4 and support recursive bisection as the canonical strategy for the Districting Integrity Act.

References

George Karypis and Vipin Kumar. A fast and high quality multilevel scheme for partitioning irregular graphs. *SIAM Journal on Scientific Computing*, 20(1):359–392, 1998.